On the Nature of Mass: Complexity Generation and Spacetime Curvature

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Abstract

We show that the radial component of the exterior Schwarzschild metric can be derived exactly from one central hypothesis and four supporting assumptions, without invoking Einstein's equations. The central hypothesis is that mass is a rate of irreversible information generation: an isolated mass M of pressureless dust produces classical information at rate $dC/d\tau = \alpha Mc^2/\hbar$, where α is a single free parameter. Extensions to matter with pressure, shear, or momentum flux are addressed in the companion paper (Paper II). Four assumptions govern how this information is geometrically accommodated: (1) information propagates at the speed of light; (2) the vacuum near matter operates at holographic saturation; (3) each nat of information requires a minimum of four Planck areas (Bekenstein); and (4) nature is parsimonious. The free parameter α is fixed without reference to the Schwarzschild metric by requiring that the information in transit between the source and any bounding sphere saturate the Bekenstein bound exactly — a consistency condition that uniquely selects $\alpha = 2\pi$. With α so fixed, the radial metric component $g_{rr} = (1-r_S/r)^{-1}$ follows algebraically as a prediction. Matching to the known Schwarzschild solution then provides confirmation rather than definition. We further present a well-motivated argument for the temporal component g_{tt}, identifying one step as a conjecture open to community scrutiny; an independent derivation of g_{tt} appears in the companion paper via the Einstein equations, themselves derived from Screen Theory premises. The argument applies equally to isolated masses of any scale. A cosmological motivation for the holographic saturation assumption, based on an Olbers-style integration of information flux from all matter in the observable universe, is developed in Appendix A.

1 Introduction

1.1 The Central Claim

Mass is ordinarily defined operationally: it measures resistance to acceleration and determines the strength of gravitational interaction. In this paper we propose a more fundamental characterization.

Mass is the rate at which a system generates irreversible classical information.

Physics knows how mass behaves: it resists acceleration, sources gravity, couples to the Higgs field. It does not know what mass *is*. The Higgs mechanism specifies the origin of the mass parameter in the Standard Model Lagrangian without addressing its geometric meaning. General relativity treats mass-energy as a source term in the field equations without asking what it is doing to produce curvature. The present paper proposes that curvature is not a response to mass but the geometric record of mass's ongoing activity: the irreversible narrowing of the universe's set of possible futures, at a rate proportional to the mass-energy present.

Spacetime curvature is the geometric accommodation required by this ongoing process. More precisely, we propose the hypothesis

$$\frac{dC}{d\tau} = \alpha \frac{Mc^2}{\hbar}, \tag{1}$$

where C denotes the amount of irreducible classical information generated by an isolated mass M, and α is a single free parameter fixed in Section 3 by the Bekenstein bound, without reference to the Schwarzschild metric. In the derivation below we show that the radial component of the exterior Schwarzschild metric follows directly from this hypothesis together with four physical assumptions governing how information is geometrically accommodated; the Schwarzschild result then constitutes a prediction confirmed by matching. Hypothesis (1) is stated for pressureless dust; extensions to matter with pressure, shear, or momentum flux are more complex and are addressed in the companion paper (Paper II). Pressureless dust isolates the mass-energy density contribution to the stress-energy tensor cleanly; pressure and momentum flux introduce additional terms in $T_{\mu\nu}$ that modify the information generation rate in ways requiring the full treatment of Paper II.

In this framework classical information is not conserved but increases monotonically through irreversible physical processes. The accumulation of such information defines a natural informational arrow of time. Spacetime geometry responds not merely to energy but to the irreversible narrowing of the set of physically possible futures.

The quantity C denotes the physical complexity of the classical record: the total surprisal $-\ln p(\text{history})$ of the complete sequence of irreversible outcomes. For locally uncorrelated R-events this reduces to the cumulative surprisal

$$C = \sum_i \big(-\ln p_i\big), \tag{2}$$

where p_i is the probability assigned by quantum mechanics to the i-th R-process outcome prior to its occurrence. This sum provides an operational lower bound on the full physical complexity; equality holds when R-events are statistically independent, which is justified by the definition of objective state reduction [6]. The full quantity coincides with the Kolmogorov complexity of the classical record up to an additive constant bounded by the description length of the laws of physics [4]. The derivation requires only the local rate $dC/d\tau$ under the uncorrelated assumption; no computability is assumed, and the lower-bound character of the sum does not affect the derivation. Unitary quantum evolution contributes zero to C: each step of a unitary evolution has $p_i = 1$, contributing zero surprisal. Only irreversible transitions — where $p_i < 1$ reflects genuine quantum indeterminacy resolved into a definite outcome — introduce new terms to the sum.

The argument originates in a simple observation: $E = hf$ and $E = Mc^2$ share a left-hand side. Setting them equal gives every isolated mass a characteristic frequency $f = Mc^2/h$ — the Compton frequency — and raises a concrete question: what is the mass doing at that frequency? The answer proposed here is that it behaves as a Shannon source [7], producing irreversible classical information at that rate. The holographic principle then supplies the geometric link: information requires physical expression as boundary area, and since it propagates outward at c, area accumulates in the surrounding region. The key physical insight is that information generated near the mass spends time proportional to r/c *in transit* before reaching a bounding surface at radius r. More area accumulates per unit radius closer to the source, giving gravity its $1/r$ character and producing the Schwarzschild radial metric.

A note on terminology: throughout this paper we use *isolated mass* rather than *particle*. The argument applies equally to a proton, a neutron star, or the observable universe. The Compton scale $\lambda_C = \hbar/Mc$ is the relevant length scale in each case, regardless of internal structure.

A note on the microscopic picture. The present paper treats mass as a rate of irreversible information generation without specifying the microscopic mechanism responsible. Companion papers develop a concrete model in which this generation arises from the interaction of mass with vacuum information threads — topological objects whose density and crossing structure constitute the local spacetime geometry. The kinematic derivation here does not depend on that picture, but the thread model provides a geometric interpretation of each step and is offered as motivation for readers seeking a more concrete substrate.

1.2 Irreversible Processes and Information Generation

Quantum dynamics contains two qualitatively different forms of evolution [6]: reversible unitary evolution (U-process), which preserves information content, and irreversible state reduction (R-process), which produces definite classical outcomes and thereby generates new classical information.

A simple example illustrates the principle. Consider a 50/50 beam splitter illuminated by single photons at a rate of 10^9 per second. Each photon is detected in either the transmitted or reflected arm, producing a binary sequence of outcomes. Because the events are statistically independent, this sequence contains roughly 10^9 bits of algorithmically incompressible classical information after one second. The mass equivalent implied by hypothesis (1) is

$$M = \frac{\hbar}{2\pi c^2}\frac{dC}{d\tau} = \frac{1.055 \times 10^{-34}}{2\pi \times 9 \times 10^{16}} \times 10^9 \approx 1.86 \times 10^{-43}\ \text{kg}. \tag{3}$$

The geometric effect is real — the surrounding area genuinely increases by $4\ell_P^2$ per nat generated — but undetectable at this rate. The example illustrates the universal mechanism operating at a scale far below the mass-vacuum interaction that dominates the curvature budget. The framework asserts that matter continuously drives analogous processes through its interaction with vacuum degrees of freedom at a rate set by the Compton frequency. The microscopic mechanism is not specified here and remains an open question addressed in companion papers.

A potential confusion deserves direct address. The complexity generation rate $dC/d\tau$ for one kilogram of matter is vastly larger than the thermodynamic entropy of the same mass ($\sim 10^{25}$ nats for a kilogram of gas at standard conditions). These are different ledgers. Thermodynamic entropy counts the arrangement of matter's own microscopic degrees of freedom. The complexity generation rate counts the ongoing output of the matter-vacuum interaction — information flowing into the structure of the surrounding vacuum, not into the internal state of the matter. The large ratio is a consistency requirement rather than a liability: the derivation produces the correct Schwarzschild geometry precisely because the generation rate is set by mass-energy (via $\hbar$ and $\alpha = 2\pi$), not by the thermodynamic state of the matter. A rate comparable to thermodynamic entropy would produce no measurable curvature.

1.3 Relation to Prior Work

The identification of quantum state reduction as the source of irreversibility follows Penrose [6, 3], who distinguished unitary from irreversible quantum evolution and proposed that R-processes are connected to spacetime geometry. Our framework inverts Penrose's causal direction: where Penrose proposed that geometry *drives* state reduction, we find that irreversible transitions *produce* geometry. The R-process is primary; curvature is its accommodation. We remain agnostic about the physical mechanism responsible for the transition itself.

The holographic principle — that the information content of a region is bounded by its boundary area — provides the bridge from information generation to geometry. Developed by 't Hooft [9] and Susskind (see Other References) and given precise form by Bekenstein's bound [1, 2], it is the second pillar of our argument. That entanglement entropy in quantum field theory scales with boundary area rather than volume [8] provides concrete evidence that area-based information structure is a natural feature of quantum spacetime, independently supporting assumption A2 below.

Several previous approaches have explored connections between gravity and information. The present framework differs from thermodynamic derivations of gravitational dynamics in a key respect: rather than treating gravity as an emergent entropic phenomenon, we identify mass itself as a rate of irreversible information generation and interpret curvature as the kinematic accommodation of that production. The companion papers develop this distinction in detail.

1.4 Hypothesis and Assumptions

The derivation rests on one central hypothesis and four assumptions.

H (Generation hypothesis). An isolated mass M of pressureless dust generates irreversible classical information at rate $dC/d\tau = \alpha Mc^2/\hbar$, where α is a single free parameter. Section 3 establishes $\alpha = 2\pi$ from the Bekenstein bound alone, without reference to the Schwarzschild metric. The derivation of g_{rr} in Section 2 then constitutes a prediction; matching to the known Schwarzschild solution provides confirmation.

A1 (Propagation). Information propagates at the speed of light. Classical information is

encoded in physical states whose propagation is bounded by c; R-process outputs cannot be transmitted superluminally without violating relativistic causality.

A2 (Saturation). The vacuum near matter operates at holographic saturation: newly generated information requires a corresponding increase in bounding area. A2 is an assumption of this framework; Srednicki [8] provides independent theoretical support from the area scaling of entanglement entropy in quantum field theory, and Appendix A provides an independent quantitative motivation from an Olbers-style integration of cosmological information flux. We note explicitly that the successful derivation of the Schwarzschild metric in Section 2 provides a posteriori consistency but not independent confirmation of A2: a derivation that uses A2 cannot serve as evidence for A2. The strongest independent motivation for A2 is the cosmological argument of Appendix A, which gives holographic saturation to within a factor of 3 under the Einstein–de Sitter approximation; the exact calculation for the concordance cosmology is developed in Paper III.

A3 (Bekenstein bound). Each nat of information requires a minimum area of $4\ell_P^2$, where $\ell_P = \sqrt{G\hbar/c^3}$.

A4 (Minimality). Nature produces no excess area beyond what the information requires.

1.5 Departures from Standard Frameworks

The central hypothesis has several consequences that depart from standard physical assumptions. We state them plainly here so that the reader can assess them explicitly rather than encounter them as surprises.

Quantum evolution is not strictly unitary. R-processes are genuinely irreversible, not merely apparent irreversibility arising from coarse-graining or loss of phase information. The framework treats state reduction as a primitive, not as something to be derived from or reconciled with unitary evolution. This is a deliberate departure from the Everettian programme; it follows Penrose's insistence that R-processes are physically real, though with the causal direction reversed.

A deeper geometric reason underlies this departure. Unitary evolution preserves the inner product $\langle\psi|\phi\rangle$, which presupposes a fixed measure and hence a fixed metric. The Ricci tensor $R_{\mu\nu}$, by contrast, is the geometric record of irreversible area-growth events and requires a dynamic metric. These two requirements are mutually inconsistent. By Gauss's *Theorema Egregium*, a surface of nonzero Gaussian curvature admits no isometric embedding in flat space: the surface cannot be flattened without tearing. Demanding that R-processes be reducible to U-processes is demanding that a curved surface be globally flattenable — possible in local normal coordinates, impossible globally. The ultraviolet divergences of perturbative quantum gravity are the precise mathematical expression of this geometric incompatibility: they are the infinities generated by the attempt to impose a flat Hilbert-space inner product on a curved-spacetime dynamics. This incompatibility has been established rigorously in the mathematical physics literature: quantum field dynamics on generic curved spacetimes is not unitarily implementable (see Related Works). The algebraic approach to QFT was developed in part to accommodate this failure by abandoning the Hilbert space apparatus. The present framework takes a different lesson from the same

result: rather than preserving unitarity by modifying the formalism, we accept that R-processes are genuinely non-unitary and identify curvature as their geometric record. The measurement problem is not a problem of interpretation. It is a geometry problem.

Information is not conserved in the sense of a fixed total. Each R-process produces a classical, objective outcome: a settled fact added permanently to the universe's record. In the sense of a museum conservator, information is never lost — settled outcomes are not undone — but the collection is always growing. The total classical information content of the universe increases monotonically. This monotonic accumulation is the physical origin of the arrow of time in the present framework. Irreversibility deepens in stages: a measurement outcome is irreversible in the practical sense (phase information gone); information scrambled with many degrees of freedom is irreversible in the thermodynamic sense; but the deepest form is causal — information crossing a horizon is placed permanently beyond the reach of every future observer within the accessible universe. Horizon-crossing is the universe's irrevocable commitment.

CPT symmetry is broken at the level of fundamental dynamics. If R-processes are genuinely time-asymmetric, CPT violation is not merely a thermodynamic artifact but a feature of the underlying physics. The expected magnitude, suppressed by approximately 1/(age of universe) $\approx 10^{-17}$–10^{-18} s^{-1}, corresponds to an energy scale of order 10^{-42} GeV — roughly 24 orders of magnitude below the current best experimental bounds from neutral kaon systems ($\sim 10^{-18}$ GeV; see Other References). The prediction is therefore consistent with all current experimental results and not accessible to near-term experiments. Whether the suppression is uniform or concentrates near compact masses where the local generation rate is high is a question for companion papers and future experiment.

Spacetime curvature is kinematic rather than dynamical. In standard general relativity, curvature is governed by field equations: it is the dynamical response of spacetime to an energy-momentum source. In the present framework, curvature is a kinematic consequence of information-theoretic constraints — the geometric shape that information generation necessarily imposes on the surrounding spacetime, in the same way that Lorentz contraction is a kinematic consequence of the invariance of c rather than a dynamical force. Whether this reframing has observable consequences beyond reproducing the standard results is a question the companion papers begin to address.

1.6 Scope of the Present Paper

The aim of the present paper is limited. We show that the radial component of the exterior Schwarzschild metric follows directly from H and A1–A4 without invoking Einstein's field equations. An argument for the temporal component is presented as a conjecture open to further scrutiny. Questions concerning the microscopic origin of the information generation process, the derivation of the Einstein equations, and cosmological information balance are addressed in companion papers.

2 Derivation of the Exterior Schwarzschild Metric

We derive the exterior Schwarzschild metric from hypothesis H and assumptions A1–A4, without invoking Einstein's field equations.

2.1 Area Response to Information Growth

The derivation rests on the following chain of reasoning. Classical information is generated through irreversible quantum state reductions. Because such reductions are non-unitary, classical information increases monotonically, providing a natural informational arrow of time. Information contained within a spatial region is subject to the Bekenstein–Hawking bound: the maximum entropy associated with a region is proportional to the area of its bounding surface,

$$S_{\max} = \frac{A}{4\ell_P^2}. \tag{4}$$

If classical information within a region increases through irreversible state reductions, the holographic bound implies that the bounding area must increase accordingly. This requirement links information growth directly to geometry.

A preliminary calculation shows that the ratio of holographic information to holographic capacity yields the exact Schwarzschild parameter. The Bekenstein bound for mass M within radius r (in natural units $\hbar = c = 1$) is

$$S_{\text{Bekenstein}} = 2\pi M r.$$

The holographic capacity of the bounding sphere of radius r (with $\ell_P^2 = G$ in natural units) is

$$S_{\max} = \frac{4\pi r^2}{4G} = \frac{\pi r^2}{G}.$$

The fractional saturation is therefore, exactly,

$$\frac{S_{\text{Bekenstein}}}{S_{\max}} = \frac{2\pi M r}{\pi r^2/G} = \frac{2GM}{r} = \frac{r_S}{r},$$

where $r_S = 2GM/c^2$ is the Schwarzschild radius. This is not a scaling result: the numerical prefactors work out exactly, with no residual dimensionless factor. The gravitational field strength at radius r is precisely the degree to which the region's informational content saturates its holographic capacity. It is exactly this ratio r_S/r that controls every deviation from flat space in both components of the Schwarzschild metric: $g_{tt} = -(1 - r_S/r)$ and $g_{rr} = (1 - r_S/r)^{-1}$.

For a general static spherically symmetric line element [10]

$$ds^2 = -A(r)\,dt^2 + B(r)\,dr^2 + r^2\,d\Omega^2, \tag{5}$$

the radial metric coefficient $B(r)$ controls the physical length associated with a coordinate radial displacement. The derivation in Sections 2.3 and 2.4 makes the informational origin of $B(r)$ precise.

2.2 Setup and Coordinate Conventions

Consider a spherically symmetric, stationary mass M of pressureless dust. We work in the exterior vacuum region $r > R$, where R is the radius of the matter distribution, and the geometry approaches flat Minkowski spacetime as $r \to \infty$. We adopt Schwarzschild coordinates (t, r, θ, ϕ) [10]. The Schwarzschild radius is

$$r_S = \frac{2GM}{c^2}. \tag{6}$$

This is a geometric length scale; it does not imply the object is a black hole. We proceed in two steps: g_{rr} from area excess (Section 2.3), then the argument for g_{tt} (Section 2.4).

2.3 The Radial Component: Area Excess from Information in Transit

By H, the central mass generates information at rate

$$\frac{dC}{d\tau} = \alpha \frac{Mc^2}{\hbar}. \tag{7}$$

By A1, information propagates outward at c. In the stationary configuration, the information currently in transit between the source and a sphere at coordinate radius r was generated during the preceding proper time interval r/c, giving

$$\Delta C = \frac{dC}{d\tau} \cdot \frac{r}{c} = \frac{\alpha M c r}{\hbar}. \tag{8}$$

By A2 (saturation), this information requires geometric expression as area. By A3 (Bekenstein), each nat requires $4\ell_P^2$ of area. By A4 (minimality), the area excess is exactly

$$\Delta A = 4\ell_P^2 \, \Delta C = 4\frac{G\hbar}{c^3} \cdot \frac{\alpha M c r}{\hbar} = \frac{4\alpha G M r}{c^2}, \tag{9}$$

which simplifies to

$$\Delta A = 2\alpha \, r_S r. \tag{10}$$

We define the radial coordinate r as the areal coordinate: the physical area of a sphere at coordinate r is $A_{\text{phys}} = 4\pi r^2$. The area that sphere would have absent the mass's information contribution is

$$A_{\text{unstretched}} = 4\pi r^2 - 2\alpha r_S r. \tag{11}$$

The radial metric component is the ratio of physical to unstretched area:

$$g_{rr} = \frac{4\pi r^2}{4\pi r^2 - 2\alpha r_S r} = \frac{1}{1 - \alpha r_S / 2\pi r}. \tag{12}$$

The area contribution from information in transit grows linearly with r — exactly the profile required for a $1/r$ gravitational potential. Section 3 establishes $\alpha = 2\pi$ from the Bekenstein bound alone, without reference to the Schwarzschild metric. Substituting $\alpha = 2\pi$:

$$g_{rr} = \frac{1}{1 - r_S/r}. \tag{13}$$

This is the radial component of the Schwarzschild metric, derived from information accounting alone. Each step follows from H and A1–A4 in sequence. No field equations, curvature tensors, or prior assumptions about geometry are required. Matching to the known Schwarzschild metric $g_{rr} = (1 - r_S/r)^{-1}$ confirms the result; Section 3 shows that the Bekenstein bound independently requires $\alpha = 2\pi$, so this matching is confirmation of a prediction rather than the definition of a fitted parameter.

2.4 The Temporal Component: A Conjecture and an Invitation

The derivation of g_{rr} is kinematic: it follows from counting information in transit. The temporal component g_{tt} requires a further argument. We present the strongest argument we have, identify the step that remains a conjecture, and invite community scrutiny. An independent derivation of g_{tt} via the Einstein equations appears in Paper II.

The configuration is stationary, so the spacetime admits a timelike Killing vector $\partial/\partial t$. The energy of any null ray — the carrier of information — is conserved along each trajectory in the exterior:

$$E = -g_{tt}\frac{dt}{d\lambda} = \text{const along each null ray.} \tag{14}$$

In the source-free exterior, no information is generated. The integrated information flux through any concentric sphere is therefore constant:

$$\Phi(r) = \text{const} \qquad \text{for all } r > R. \tag{15}$$

The most general static spherically symmetric metric takes the form [10]

$$ds^2 = -A(r)\,dt^2 + B(r)\,dr^2 + r^2\,d\Omega^2, \tag{16}$$

with $B(r) = (1 - r_S/r)^{-1}$ already determined by Section 2.3. We write

$$A(r) = e^{2\Phi(r)}\left(1 - \frac{r_S}{r}\right) \tag{17}$$

for a free function $\Phi(r)$. Asymptotic flatness requires $\Phi(r) \to 0$ as $r \to \infty$; the general expansion is $\Phi(r) = \beta/r + \gamma/r^2 + \cdots$ Each nonzero coefficient introduces a free parameter that must be sourced by some physical field. In the present setup — vacuum exterior, single length scale r_S already fixed, no additional fields — no such source is present.

Conjecture: $\Phi(r) = 0$ identically.

By A4 (minimality), no geometric structure is produced beyond what the information constraints require. A nonzero $\Phi(r)$ would modify the proper time measured by stationary clocks at radius r, changing the rate at which information generated near the source is red- or blue-shifted as it propagates outward. This modified propagation rate would require additional area accommodation beyond what the information in transit already demands — area sourced by nothing in the physical setup, violating A4. Therefore $g_{tt} = -(1 - r_S/r)$.

Where the argument needs scrutiny. A4 is doing substantial work here. In GR, $\Phi = 0$ follows from the vacuum Einstein equations via Birkhoff's theorem — a rigorous result. We reach the same conclusion by a different route. The open problem is to show rigorously that A4 excludes all unsourced free parameters in the temporal component, without implicitly assuming the conclusion. We present this as a conjecture; the companion paper (Paper II) provides an independent derivation that may serve as confirmation pending a direct proof.

2.5 The Exterior Schwarzschild Metric

If the conjecture of Section 2.4 holds, the complete line element in the exterior region $r > R$ is

$$ds^2 = -\left(1 - \frac{r_S}{r}\right) dt^2 + \left(1 - \frac{r_S}{r}\right)^{-1} dr^2 + r^2\left(d\theta^2 + \sin^2\theta \, d\phi^2\right), \tag{18}$$

with $r_S = 2GM/c^2$ and $\alpha = 2\pi$. The radial component is exact. The temporal component depends on the conjecture of Section 2.4. Together they constitute the exterior Schwarzschild metric.

3 Fixing the Free Parameter: $\alpha = 2\pi$ from the Bekenstein Bound

The hypothesis H contains one free parameter α. We now show that this parameter is fixed uniquely by the Bekenstein bound, without invoking the Schwarzschild metric. The derivation of g_{rr} in Section 2 is therefore a prediction, not a fitting exercise; the Schwarzschild matching confirms it.

3.1 The Bekenstein Bound

The Bekenstein bound limits the information content of a region of radius r containing energy $E = Mc^2$. In natural units ($\hbar = c = 1$):

$$K_{\max} = 2\pi M r. \tag{19}$$

The factor 2π appears in Bekenstein's original derivation, reflecting the relationship between energy, radius, and holographic capacity.

3.2 Information in Transit with Free Parameter

From Section 2.3, the information in transit between the source and a sphere at radius r (in natural units $\hbar = c = 1$) is, for general α:

$$\Delta C = \alpha M r. \tag{20}$$

3.3 Fixing α by Saturation

We require that the information in transit saturate the Bekenstein bound at every radius:

$$\Delta C = K_{\max}. \tag{21}$$

Substituting equations (19) and (20):

$$\alpha M r = 2\pi M r \quad \text{for all } r, \tag{22}$$

which requires

$$\alpha = 2\pi. \tag{23}$$

This fixes the free parameter of hypothesis H from the Bekenstein bound alone, without reference to the Schwarzschild metric. The exact factor of 2π is not a fitted value; it is the only value consistent with holographic saturation at every radius.

With $\alpha = 2\pi$ established, the information in transit is:

$$\Delta C = 2\pi M r, \tag{24}$$

and the radial metric component derived in Section 2.3 becomes the prediction

$$g_{rr} = \frac{1}{1 - r_S/r}, \tag{25}$$

confirmed by matching to the known Schwarzschild solution.

Saturation as a governing principle. Mass generates information at exactly the maximum rate compatible with the geometric capacity of its surroundings. This "just-in-time" character ensures consistency without fine-tuning.

Compton scale validation. The equality holds at all radii, including the Compton wavelength $\lambda_C = \hbar/Mc$, providing a natural consistency check at the quantum scale.

3.4 Summary

The requirement that information in transit saturates the Bekenstein bound at every radius fixes $\alpha = 2\pi$ uniquely, with no reference to the Schwarzschild metric. With $\alpha = 2\pi$, the rate $dC/d\tau = 2\pi Mc^2/\hbar$: reproduces the Schwarzschild radial metric (Section 2.3) as a prediction; saturates the Bekenstein bound at every radius (this section); and yields the factor 2π as an inevitable consistency requirement rather than a free choice. The Schwarzschild matching of Section 2.3 confirms this prediction.

4 Discussion and Conclusions

4.1 Summary of Results

We have derived the radial component of the exterior Schwarzschild metric from one hypothesis (H) and four assumptions (A1–A4), without invoking Einstein's field equations. The free parameter α in H is fixed to $\alpha = 2\pi$ by the Bekenstein bound alone, without reference to the Schwarzschild metric (Section 3); the derivation of g_{rr} then constitutes a prediction, and matching to the known Schwarzschild solution confirms it. The radial component g_{rr} is exact. The temporal component g_{tt} rests on a well-motivated conjecture identified in Section 2.4; an independent derivation appears in Paper II.

4.2 Extensions in Companion Papers

The following topics are developed in companion papers.

Einstein equations (Paper II). The full Einstein field equations for vacuum and pressureless dust, derived from Screen Theory premises. g_{tt} is recovered here as a theorem rather than a conjecture.

Cosmology (Paper III). Global information balance, Machian origin of mass-energy, and the cosmological constant reinterpreted in information-theoretic terms.

Horizons and black holes (Paper IV). Behavior at and inside $r = r_S$, information export rates, and the Page curve in information-theoretic terms.

4.3 Conceptual Implications

The derivation suggests a fundamental reinterpretation. Mass is not a source of curvature in the sense of a cause producing an effect; it *is* a rate of information generation, and curvature is the geometric shape that rate imposes on its surroundings. This makes precise Wheeler's [11] "it from bit" vision: geometry (*that*) from information generation (*nat*).

The causal order is inverted relative to standard GR. In GR, mass-energy is a source term in dynamical field equations and curvature is the solution. Here, irreversible information generation is primary and geometry is its kinematic accommodation. The Compton frequency $Mc^2/\hbar$, typically interpreted as a quantum oscillation frequency, acquires new significance as the characteristic rate of irreversible constraint production.

Newton's gravitational constant reinterpreted. From the derivation chain, G can be written as

$$G = \frac{\ell_P^2 c^3}{\hbar},$$

where $4\ell_P^2$ converts area to information (A3) and $\hbar$ converts action to information. G is therefore not a fundamental coupling constant but an *exchange rate* between Planck area and quantum of action. The factor c^3 adjusts units only. G's apparent smallness is not a mystery requiring explanation but a consequence of the ratio between two independently motivated fundamental quantities.

Proper radial length and holographic saturation. Once g_{rr} is in hand, the ratio of coordinate displacement to proper radial length follows immediately:

$$\frac{dr}{d\ell} = \frac{1}{\sqrt{g_{rr}}} = \sqrt{1 - \frac{r_S}{r}} = \sqrt{1 - \frac{S_{\text{Bekenstein}}}{S_{\max}}}.$$

The ratio of coordinate to proper radial length is the square root of the *unsaturated* holographic fraction. At $r = r_S$, saturation is complete: $dr/d\ell \to 0$, and an infinite proper distance separates any exterior point from the horizon. The horizon is the surface of complete holographic saturation, and the divergence of g_{rr} there is not a coordinate artifact but a direct geometric expression of that saturation.

4.4 An Open Problem and an Invitation

The derivation of g_{rr} is rigorous within the stated assumptions. The argument for g_{tt} rests on a step — minimality excluding unsourced free parameters — that we present as a conjecture. A rigorous proof of this step, a counterexample forcing modification of the framework, or an alternative derivation closing the gap, would each be a valuable contribution. The companion paper (Paper II) provides an independent route to g_{tt} via the Einstein equations, which may serve as confirmation of the conjecture pending a direct proof.

4.5 Forward-Looking Conjectures

Microscopic thread structure. The present paper derives the Schwarzschild geometry from information accounting without specifying the microscopic objects that carry and generate the information. Companion papers develop a model in which irreversible information generation arises from the interaction of mass with vacuum information threads — topological objects whose crossing density and biographical structure constitute local spacetime geometry. In that picture, mass is a persistent pattern of thread crossings at the Compton rate, and curvature is the area accumulated by threads propagating outward from those crossings. The thread model provides a concrete realization of hypothesis H and a geometric interpretation of each step in the derivation above.

Shape independence. Bekenstein's bound assumes nothing about interior mass distribution. The present argument may inherit this insensitivity; the spherically symmetric case may be the simplest instance of something more general.

Screen overlap and Newton's law. Newtonian gravity and its relativistic extension may admit a unified interpretation via holographic screen overlap between masses, with the nonlinearity of Einstein's equations corresponding to the breakdown of the small-overlap approximation.

Cosmological information accumulation. The accumulation of classical information on cosmic horizons — the irreversible deposit of settled outcomes beyond the reach of all future observers — may drive large-scale spacetime dynamics. This connection, including the Machian origin of mass-energy and the derivation of $E = Mc^2$ from Screen Theory premises, is developed in Paper III.

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Citations

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Other References

A note on methodology. The following works were identified as closely related to the present paper by an AI assistant (Claude, Anthropic, March 2026), on the basis of semantic content matching against the physics literature, or are works the authors are aware of but have not yet read. The authors cannot vouch for the accuracy of the AI's assessments or for the correctness of specific claims attributed to these works.

We list them here because the standard convention of citing only works one has actually read creates a systematic gap in the scholarly record. An AI can identify semantic neighbors in the literature rapidly and reliably; human authors are better placed to evaluate whether a work is correct and relevant in depth. Separating these two tasks — neighbor identification from depth evaluation — serves science better than conflating them under a single citation convention.

We propose that **Other References** sections, explicitly identified as AI-assisted or colleague-assisted rather than author-read, constitute a distinct and honest category of scholarly attribution. Such sections provide readers with a navigable neighborhood of the paper's ideas without misrepresenting the authors' familiarity with that neighborhood. We invite the community to adopt this convention.

Readers who find errors in the AI assessments, or who are aware of additional closely related works, are encouraged to contact the authors.

1. Arageorgis, A., Earman, J., and Ruetsche, L. (2002). Weyling the time away: the non-unitary implementability of quantum field dynamics on curved spacetime. *Studies in the History and Philosophy of Modern Physics*, 33(2), 151–184. *[AI assessment: establishes rigorously that QFT dynamics on generic curved spacetimes is not unitarily implementable — the dilemma identified here is directly relevant to the departures from standard frameworks discussed in Section 1.5. The present framework's response to this dilemma differs from the algebraic approach advocated in that paper.]*

2. Verlinde, E. (2011). On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011(4), 29. *[AI assessment: derives gravitational dynamics from entropic considerations, sharing the broad programme of connecting gravity to information. The present framework differs in identifying mass (not gravity) as the fundamental information-theoretic quantity, and in treating curvature as kinematic rather than entropic. The precise distinction is developed in the companion papers.]*

3. Jacobson, T. (1995). Thermodynamics of spacetime: the Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263. *[AI assessment: derives the Einstein equations from thermodynamic considerations applied to local Rindler horizons. Related in spirit to the present derivation but different in both starting point and method: Jacobson assumes the Clausius relation as fundamental, whereas the present framework derives geometry from information generation rates.]*

4. Bilson-Thompson, S. O. (2005). A topological model of composite preons. arXiv:hep-ph/0503213. *[AI assessment: represents Standard Model first-generation fermions and gauge bosons as specific braids of three ribbons — directly relevant to the thread/braid picture developed in the companion papers and flagged as an open correspondence problem in the present programme.]*

5. Penrose, R. (1971). Angular momentum: an approach to combinatorial space-time. In Bastin, T. (ed.), *Quantum Theory and Beyond*. Cambridge University Press. *[AI assessment: the original twistor/spin-network paper proposing that spacetime geometry should emerge from discrete combinatorial structure — the intellectual ancestor of approaches (including the present one) that treat spacetime points as derived rather than primitive objects.]*

6. Navas, S. et al. (Particle Data Group) (2024). *Review of Particle Physics. Physical Review D*, 110, 030001. *[Not yet read by the authors. Cited in Section 1.5 for the experimental bound on CPT violation from neutral kaon systems.]*

7. Sciama, D. W. (1953). On the origin of inertia. *Monthly Notices of the Royal Astronomical Society*, 113, 34–42. *[Not yet read by the authors. The Olbers-style calculation in Appendix A is structurally identical to Sciama's integral for inertial origin, as noted in that appendix.]*

8. Susskind, L. (1995). The world as a hologram. *Journal of Mathematical Physics*, 36(11), 6377–6396. *[Not yet read by the authors. Co-developer of the holographic principle with 't Hooft; cited in Section 1.3 as a pillar of the argument.]*

A Cosmological Motivation for Assumption A2

Assumption A2 — that the vacuum near matter operates at holographic saturation — is the least self-evident of the four assumptions underlying the derivation. We offer here a quantitative motivation, deferring the exact calculation for the concordance cosmology to Paper III.

Intellectual context

The question of where space comes from has a long history. Leibniz argued against Newton that space has no existence independent of the relations between objects; to posit an absolute space is to posit something with no observable consequences and no causal role. Mach [5] extended this relational programme to inertia, proposing that the resistance of matter to acceleration is not a response to absolute space but to the gravitational influence of all other matter in the universe. General relativity absorbed Mach's relational spirit without fully vindicating it: the metric is relational in form, but the manifold remains a given substrate.

Sciama (see Other References) gave Mach's proposal its most precise quantitative form, showing that the sum of gravitational potential contributions $\sum GM_i/r_i c^2$ from all matter in the observable universe yields a result of order unity, suggesting that the inertia of any local object is entirely due to its interaction with distant matter. The calculation below is structurally identical to Sciama's integral (see Other References), evaluated in information-theoretic language rather than gravitational potential language. The agreement is not a coincidence; both are expressions of the same underlying physics.

The Olbers integral for holographic area

We model the universe as Einstein–de Sitter: spatially flat, matter-dominated, $\Omega = 1$, $\Lambda = 0$, with critical density $\rho_c = 3H_0^2/8\pi G$ and particle horizon $r_H = 2c/H_0$. Rather than summing light flux from distant sources as in Olbers's paradox, we sum holographic area contributions.

From Section 2.3, a mass M at distance r contributes information in transit

$$\Delta C = \frac{2\pi M c r}{\hbar}, \tag{26}$$

which by A3 requires holographic area

$$\Delta A = 4\ell_P^2\, \Delta C = \frac{8\pi G M r}{c^2}. \tag{27}$$

Integrating over concentric shells at critical density:

$$A_{\text{required}} = \int_0^{r_H} \frac{8\pi G r}{c^2}\, \rho_c\, 4\pi r^2\, dr = \frac{128\pi^2 G \rho_c}{c^2} \cdot \frac{r_H^4}{4} = 48\pi \frac{c^2}{H_0^2}. \tag{28}$$

The holographic capacity of the cosmological horizon is

$$A_{\text{horizon}} = 4\pi r_H^2 = 16\pi \frac{c^2}{H_0^2}, \tag{29}$$

giving the ratio

$$\frac{A_{\text{required}}}{A_{\text{horizon}}} = 3. \tag{30}$$

This is a factor-of-3 result with no free parameters.

By the Copernican principle this holds at every point: each location receives information flux from its own observable universe, and that flux saturates the local holographic screen

to within a factor of 3 under the Einstein–de Sitter approximation. We conjecture that the exact calculation for the concordance cosmology — incorporating dark energy and the redshift evolution of matter density — closes this gap and yields exact saturation. The factor of 3 overshoot in the EdS case is qualitatively consistent with dark energy reducing the effective matter volume relative to horizon area.

Physical interpretation

This result provides independent quantitative motivation for A2: the vacuum operates at holographic saturation not because we postulate it, but because the collective information flux from all matter in the observable universe fills the local holographic screen to capacity at every point.

It also suggests a more complete resolution to the question Leibniz and Mach were asking. Space is not a pre-existing stage. The local vacuum exists, and has the geometric structure it has, because the universe has been generating information long enough and densely enough that the past light cone of every point is saturated. Spacetime is not ordained — it is accumulated, built incrementally from the information output of all matter within causal reach.

This reframes the apparent wastefulness of cosmic voids. Every patch of apparently empty space encodes the complete history of its past light cone, back to the beginning of time: a holographic record of everything that has ever been causally accessible from that location. Matter is a light dusting of active information-generation centers embedded in a universe that is overwhelmingly composed of this accumulated record.

The Machian implication runs deeper still. The same integral that gives equation (30) is structurally identical to Sciama's calculation of inertial origin. This suggests that mass-energy itself — and ultimately $E = Mc^2$ — has a Machian, relational origin: the energy content of matter is built up from its interactions with the cosmological information field rather than being intrinsic to matter in isolation. Screen Theory may thus vindicate Mach's programme in a precise quantitative sense: both space and inertia are accumulated, relational quantities, not given substrates. This connection is developed in Paper III.

Note on placement

This appendix presents the argument at the level of a toy calculation sufficient to establish plausibility. The full derivation — with proper cosmological distance measure, redshift corrections, and exact treatment of the concordance cosmology — is reserved for Paper III, where it forms part of the central argument. This appendix may be read independently of the main text and may be updated or replaced as the Paper III calculation is completed.